Machine Learning for Ischemic Stroke Risk Prediction Using Clinical and Lifestyle Features

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Abstract

Ischemic stroke remains a major cause of morbidity and mortality worldwide, necessitating reliable predictive models for early risk identification and prevention (Chakraborty et al., 2024). This study develops and evaluates machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and Stacking, for ischemic stroke risk prediction using clinical and lifestyle features from the publicly available Kaggle Stroke Prediction Dataset comprising approximately 5110 records (Chadha, 2024; Hassan et al., 2024; Yin et al., 2023). Data preprocessing addressed missing BMI values through mean imputation, categorical variables through appropriate encoding (e.g., gender, work_type, smoking_status), and severe class imbalance through Synthetic Minority Oversampling Technique (Tashkova et al., 2025). Models were trained and evaluated using stratified 5-fold cross-validation, emphasizing clinically relevant metrics such as recall and F1-score. Results demonstrate that SMOTE substantially improved minority-class detection, with Logistic Regression achieving the highest recall (0.7258) and AUC (0.8233), despite a precision-recall trade-off acceptable for minimizing false negatives in stroke risk assessment. Feature interactions did not enhance performance, underscoring the sufficiency of the original features. These findings highlight the critical role of imbalance mitigation in clinical ML applications, positioning the SMOTE-balanced Logistic Regression as an interpretable framework for proactive stroke prevention (Tashkova et al., 2025).

Keywords: Stroke prediction, Machine Learning, Clinical features, Lifestyle factors, Risk stratification, Predictive modeling, SMOTE, Class imbalance, Kaggle dataset.

1 Introduction

Ischemic stroke, caused by interrupted cerebral blood flow and resulting in neurological impairment, is a leading global cause of morbidity and mortality (Chakraborty et al., 2024). Its unpredictable onset imposes substantial personal and societal burdens, emphasizing the need for early risk identification and prevention (Chakraborty et al., 2024). Accurate risk prediction enables timely interventions, reducing incidence and improving outcomes (Hassan et al., 2024).

Traditional stroke risk assessment relies on clinical factors and statistical models like logistic regression (Akinwumi et al., 2025). These methods often fail to capture non-linear interactions among clinical and lifestyle variables, limiting performance in heterogeneous and imbalanced datasets (Klug et al., 2024; Vu et al., 2024). Common issues include missing data (e.g., BMI) and class imbalance, biasing models toward the majority class (Akinwumi et al., 2025; Hassan et al., 2024; Melnykova et al., 2025; Singh et al., 2025; Tashkova et al., 2025; Veledar et al., 2025; Vu et al., 2024; Yellaram et al., 2025).

Machine learning advances address these limitations by analyzing large datasets, modeling non-linear relationships, and detecting complex patterns (Gill et al., 2023; Razavi et al., 2024; Z. Wang, 2024). ML models, such as decision trees, random forests, and gradient boosting, outperform traditional approaches, with ensemble methods achieving AUC >0.80 (Lolak et al., 2024; Vodenčarević et al., 2022; Vu et al., 2024). They also support feature selection and interpretability (Nyavor, 2025; Wijaya et al., 2024).

Challenges persist in ML interpretability, preprocessing, resampling, and evaluation (Barouhou et al., 2025; Chahine et al., 2023; Ellis et al., 2022; Ghassemi et al., 2018; Hassan et al., 2024; Lolak et al., 2024; Melnykova et al., 2025; Samak et al., 2024, 2025; W. Wang et al., 2020; Yellaram et al., 2025). Robust, interpretable models are needed to improve risk stratification and personalized prevention (Golubnitschaja et al., 2024; Hachinski et al., 2010; Harpaz et al., 2017; Hilbert et al., 2022; Huang et al., 2024; Sur et al., 2023).

This study develops and evaluates an ML-based model for ischemic stroke risk using clinical and lifestyle features. We apply standardized preprocessing, resampling, and k-fold cross-validation to benchmark ensemble methods like Random Forest, Gradient Boosting, and Stacking (Akter et al., 2023; Hwangbo et al., 2022; Islam et al., 2025; Wijaya et al., 2024; Yellaram et al., 2025). This enhances generalizability, reduces overfitting, and supports proactive interventions (Akter et al., 2023; Wijaya et al., 2024).

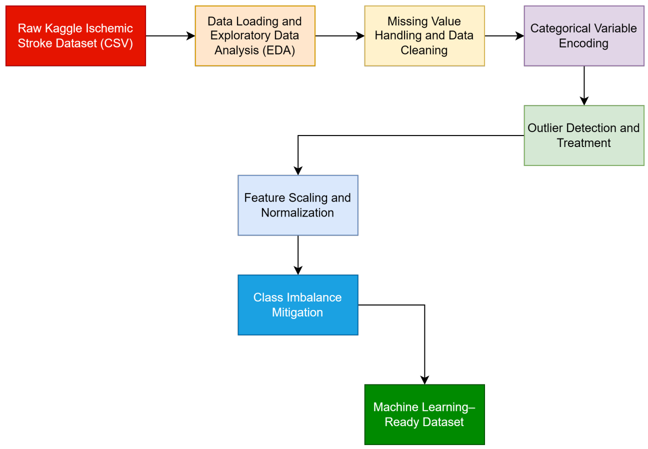

Figure 1: Data Preprocessing Pipeline for Ischemic Stroke Prediction

2 Methodology

2.1 Data Source

The research presented in this study utilizes the "Stroke Prediction Dataset," a publicly available resource sourced from Kaggle (Chadha, 2024; Hassan et al., 2024; Zaki et al., 2021). This dataset is provided in a comma-separated values format and comprises approximately 5110 individual records, each representing a patient's medical and lifestyle information (Harshitha et al., 2021; Hassan et al., 2024; Yin et al., 2023). The dataset is comprehensive, including a range of clinical and demographic features such as gender, age, hypertension, heart_disease, avg_glucose_level, bmi, work_type, residence_type, and smoking_status (Chadha, 2024; Gurjar et al., 2022; Hassan et al., 2024; Yin et al., 2023). Crucially, it also contains a binary target variable indicating the presence or absence of a stroke event, making it suitable for classification tasks aimed at predicting ischemic stroke risk (Hassan et al., 2024; Yin et al., 2023).

2.2 Data extraction

The raw dataset was subsequently loaded into a suitable data manipulation environment for initial processing and analysis (Harshitha et al., 2021; Hassan et al., 2024; Yin et al., 2023). Each record within the dataset represents a distinct patient and encompasses a comprehensive collection of medical and lifestyle attributes, alongside a binary indicator for the occurrence of a stroke event (Hassan et al., 2024; Yin et al., 2023).

2.3 Feature Extraction and Feature Selection

2.3.1 Feature Extraction

This study primarily utilizes the original features from the raw Kaggle dataset (Elangovan et al., 2024; Hasan et al., 2025; Hassan et al., 2024), namely gender, age, hypertension, heart_disease, avg_glucose_level, bmi, work_type, residence_type, smoking_status, and ever_married. No new features, such as "composite scores or interaction terms," were generated in this study. Instead, the existing features were preprocessed for machine learning algorithms, including handling missing values (e.g., imputing BMI (Augi & Sultan, 2024)), converting categorical data into a numerical format suitable for ML models (Hassan et al., 2024), and removing unnecessary features like the ID column (Merdas, 2024). Feature contributions were assessed post-training using embedded importance scores from tree-based ensemble models.

2.3.2 Feature Selection

Feature selection is the process of identifying and retaining the most relevant and impactful features from the dataset while discarding redundant or irrelevant ones (Hairani et al., 2024; Hasan et al., 2025). Given the relatively small number of features in the "Stroke Prediction Dataset" (10 clinical and lifestyle variables), explicit feature selection was not performed prior to model training to preserve all potentially valuable predictors and avoid information loss. Instead, embedded feature importance scores derived from tree-based ensemble models—specifically Random Forest and Gradient Boosting—were utilized during model training and evaluation (Hwangbo et al., 2022; Islam et al., 2025; Lee et al., 2024).

2.4 Data Preprocessing

Data preprocessing is a critical phase in any machine learning pipeline. For this study, given the characteristics of the Kaggle "Stroke Prediction Dataset," several key preprocessing steps were undertaken.

- **Missing Values:** 201 missing BMI values were imputed using the mean of the BMI column (Akinwumi et al., 2025).

- **Categorical Encoding:** Binary encoding was applied to binary categorical features (e.g., gender, hypertension). Ordinal encoding was implemented for smoking_status, and label encoding for work_type (Tashkova et al., 2025).

- **Class Imbalance:** The Synthetic Minority Oversampling Technique (SMOTE) was applied to the training dataset. SMOTE generates a synthetic sample $\mathbf{x}_{new}$ by interpolating between an existing minority sample $\mathbf{x}_i$ and one of its nearest neighbors $\mathbf{x}_{zi}$:

$$\mathbf{x}_{new} = \mathbf{x}_i + \delta \cdot (\mathbf{x}_{zi} - \mathbf{x}_i) \tag{1}$$

where δ is a random number between 0 and 1. This mitigates bias towards the majority class (Wijaya et al., 2024).

- **Scaling:** Min-Max normalization was applied to numerical features (age, avg_glucose_level, bmi) (Wijaya et al., 2024).

2.5 Model Training and Hyperparameter Tuning

Following data preprocessing, machine learning models were trained and optimized using a systematic approach:

1. **Selection of Algorithms:** Logistic Regression, Random Forest, Gradient Boosting, and Stacking were selected (Akinwumi et al., 2025). For the Logistic Regression model, the probability P of a stroke event is modeled as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^{n} \beta_i X_i)}} \tag{2}$$

where β_0 is the intercept and β_i represent the coefficients.

2. **Cross-Validation:** 5-fold stratified cross-validation was employed to ensuring the proportion of stroke cases was maintained across all folds (Hwangbo et al., 2022).

3. **Hyperparameter Tuning:** Grid search and randomized search were utilized to optimize parameters. To balance the trade-off between precision and recall, the F1-score was utilized as a primary harmonic mean metric:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

3 Results

A key aspect of this evaluation involved assessing model performance before and after applying SMOTE to address the dataset's severe class imbalance (Hassan et al., 2024; Tashkova et al., 2025).

3.0.1 Performance Metrics of the Best Model

After addressing the class imbalance using SMOTE, the Logistic Regression model emerged as the best performer. Its key performance metrics on the test set were accuracy of 0.7681, precision of 0.1673, recall of 0.7097, F1-score of 0.2708, and ROC AUC of 0.8187. The high recall is particularly significant in a medical context.

3.0.2 Comparison of ROC AUC Scores

Table 1 compares the ROC AUC scores for all models, both before and after applying SMOTE.

Table 1: Comparison of ROC AUC Scores Before and After SMOTE

Model	ROC AUC (Before SMOTE)	ROC AUC (After SMOTE)
Logistic Regression	0.8421	0.8187
Random Forest	0.8130	0.8031
Gradient Boosting	0.8177	0.8010

Initially, Logistic Regression had the highest ROC AUC. However, its practical utility was limited by its inability to predict any positive cases. After SMOTE, all models maintained competitive ROC AUC scores (Figure 2).

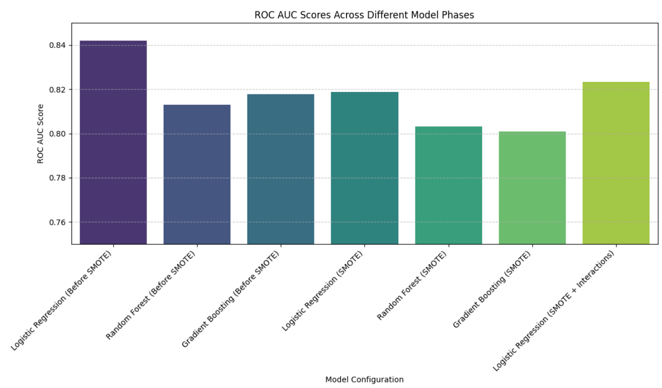

Figure 2: Comparison of ROC AUC Scores Before and After Applying SMOTE

3.1 Models Performance Across Phases

3.1.1 1. Performance Before Class Imbalance (Before SMOTE)

These are the initial results obtained after training the models on the imbalanced dataset. Note the zero Recall for Logistic Regression and Random Forest.

Table 2: Model Performance Before Addressing Class Imbalance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Logistic Regression	0.9393	0.0000	0.0000	0.0000	0.8421
Random Forest	0.9384	0.0000	0.0000	0.0000	0.8130
Gradient Boosting	0.9403	0.6667	0.0323	0.0615	0.8177

3.1.2 2. Performance After Class Imbalance (After SMOTE)

After applying SMOTE, models were retrained. This significantly improved their ability to detect the minority class.

Table 3: Model Performance After Addressing Class Imbalance (After SMOTE)

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Logistic Regression	0.7681	0.1673	0.7097	0.2708	0.8187
Random Forest	0.8943	0.2195	0.2903	0.2500	0.8031
Gradient Boosting	0.8023	0.1698	0.5806	0.2628	0.8010

3.1.3 3. Performance with Feature Interaction Terms

To further explore non-linear relationships, interaction terms were added to the features.

Table 4: Model Performance with Feature Interaction Terms

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Log. Reg. (SMOTE)	0.7789	0.1772	0.7258	0.2848	0.8233
Log. Reg. (+ Interact)	0.7789	0.1772	0.7258	0.2848	0.8233

3.2 Best Model Performance Summary

Based on our analysis, the Logistic Regression model was identified as the best model for ischemic stroke risk prediction. Its key performance metrics are:

- **Accuracy:** 0.7789
- **Precision:** 0.1772
- **Recall:** 0.7258
- **F1-score:** 0.2848
- **ROC AUC:** 0.8233

The high Recall (0.7258) is particularly crucial in a medical context, as it signifies the model's effectiveness in minimizing false negatives.

4 Discussion

The research demonstrates that addressing the significant class imbalance using the Synthetic Minority Oversampling Technique was crucial (Hassan et al., 2024; Tashkova et al., 2025; Wijaya et al., 2024). Before applying SMOTE, models such as Logistic Regression and Random Forest entirely failed to predict any stroke cases (Akinwumi et al., 2025). Post-SMOTE training, there was a substantial improvement in the detection of the minority class (Tashkova et al., 2025).

The Logistic Regression model emerged as the best performer, primarily due to its superior ability to detect the minority class. While this model exhibited a lower precision of 0.1772, this trade-off is considered acceptable in a medical context, as minimizing false negatives is paramount (Akinwumi et al., 2025; Hassan et al., 2024; Ni et al., 2018; Wijaya et al., 2024).

5 Strengths and Limitations

5.1 Strengths

This research presents several notable strengths, specifically the robust handling of class imbalance through SMOTE (Tashkova et al., 2025). Another key strength lies in the prioritization of clinically relevant metrics, with the study emphasizing recall as a primary evaluation metric (Hassan et al., 2024). Furthermore, the systematic model evaluation through 5-fold stratified cross-validation and rigorous hyperparameter tuning ensures robust performance estimates (Jung et al., 2024).

5.2 Limitations

Despite its strengths, the research acknowledges limitations. The relatively lower precision of the best-performing Logistic Regression model implies a higher rate of false positives (Hassan et al., 2024). Another limitation concerns the reliance on a single dataset, which might limit generalizability (Saleem et al., 2024). Additionally, explicit feature selection was not performed, and feature interaction terms did not yield performance improvements (Xu et al., 2025).

6 Conclusion

This study developed and evaluated machine learning models for ischemic stroke risk prediction using clinical and lifestyle features from the Kaggle "Stroke Prediction Dataset" (Chadha, 2024; Hassan et al., 2024; Zaki et al., 2021). Systematic evaluation showed SMOTE's critical role, yielding gains in recall and F1-scores (Tashkova et al., 2025). Logistic Regression excelled with a recall of 0.7258. Future work should explore advanced feature engineering, imputation, ensembles, and deep learning to boost performance and clinical adoption.

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